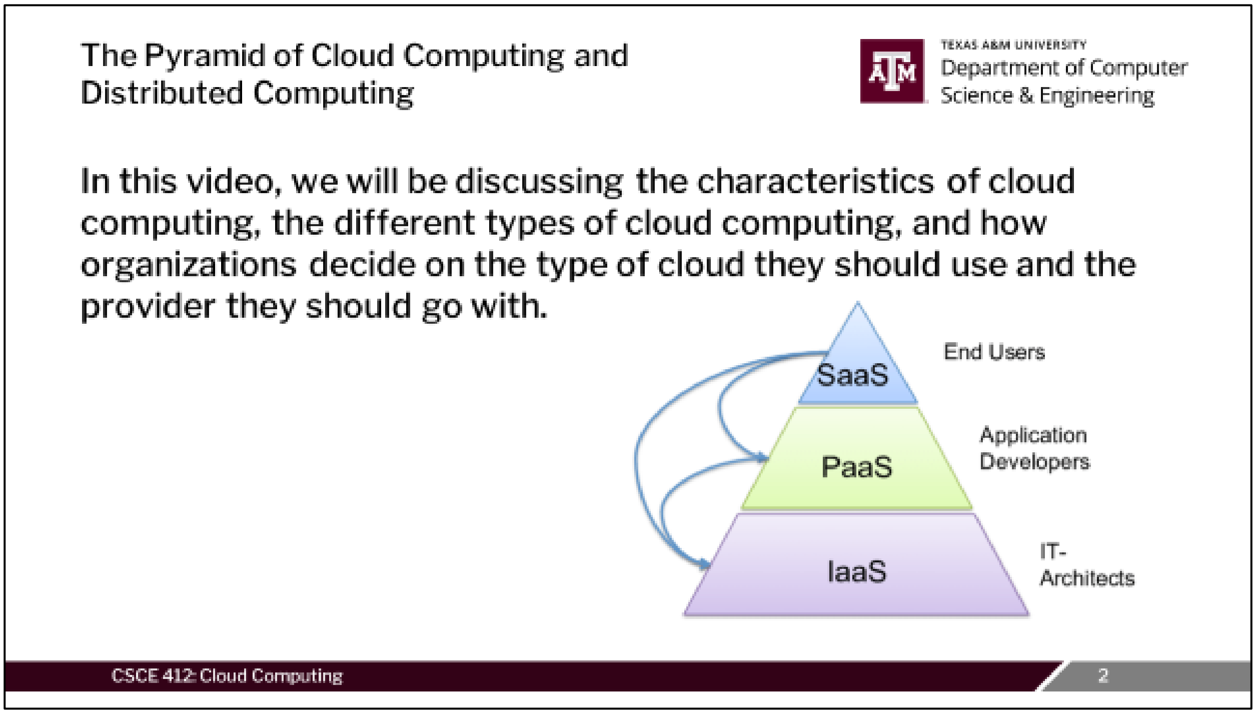
Distributed Computing Transcript

Slide 1

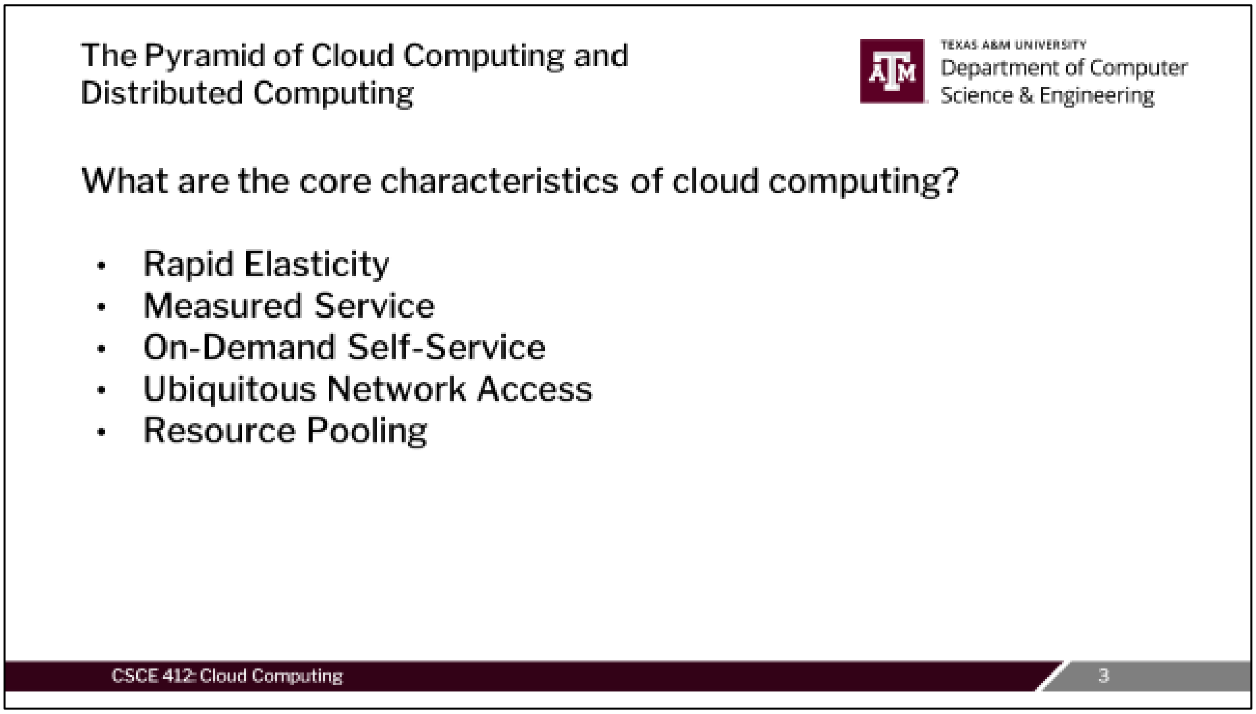
Howdy, My name is Robert Lightfoot. Welcome to this module in the CSCE 412 series, Cloud Computing.

Slide 2



In this video, we will be discussing the characteristics of cloud computing, the different types of cloud computing, and how organizations decide on the type of cloud they should use and the provider they should go with.

Slide 3



So, what are the core characteristics of cloud computing? To be considered an effective cloud provider, you should be able to provide and maintain five core characteristics.

The first is rapid elasticity and scalability, where cloud resources are rapidly provisioned and de-provisioned as demand increases and decreases. This can apply to computing power, networking capacity, storage capacity, etc. This characteristic allows for cloud computing to satisfy dynamic business needs while optimizing cost-effectiveness of services more easily. The second characteristic is measured service. Relating back to rapid elasticity and its cost optimizations, this characteristic is to meter and measure the amount of resources used by an organization to enable charge-per-use on the services. In doing so, cloud providers can charge customers with more gradients by reducing charges during low utilization periods.

Additionally, it allows the provider to monitor a customer's usage over an extended period and better predict and plan for spikes and peak utilization. The third characteristic is on-demand self-service, which allows customers to provision additional servicers without speaking directly to the cloud service provider. This is typically done by a web interface or portal that allows customers to perform self-service as they need. The fourth characteristic is ubiquitous network access, which allows for access to cloud resources via standard networking protocols, like the internet.

This also related to the provider's network capacity in terms of bandwidth and latency as it affects the overall quality of service. The fifth characteristic is resource pooling and the use of the multi-tenant model. This allows cloud providers to service multiple customers from a single physical system through the use of virtualization. Using this model, cloud providers are able to provide economically efficient services at scale. Overall, these characteristics overlap or contribute to other characteristics considerably, particularly elasticity. Using these five characteristics, cloud service providers can accommodate users of all types and sizes of needs.

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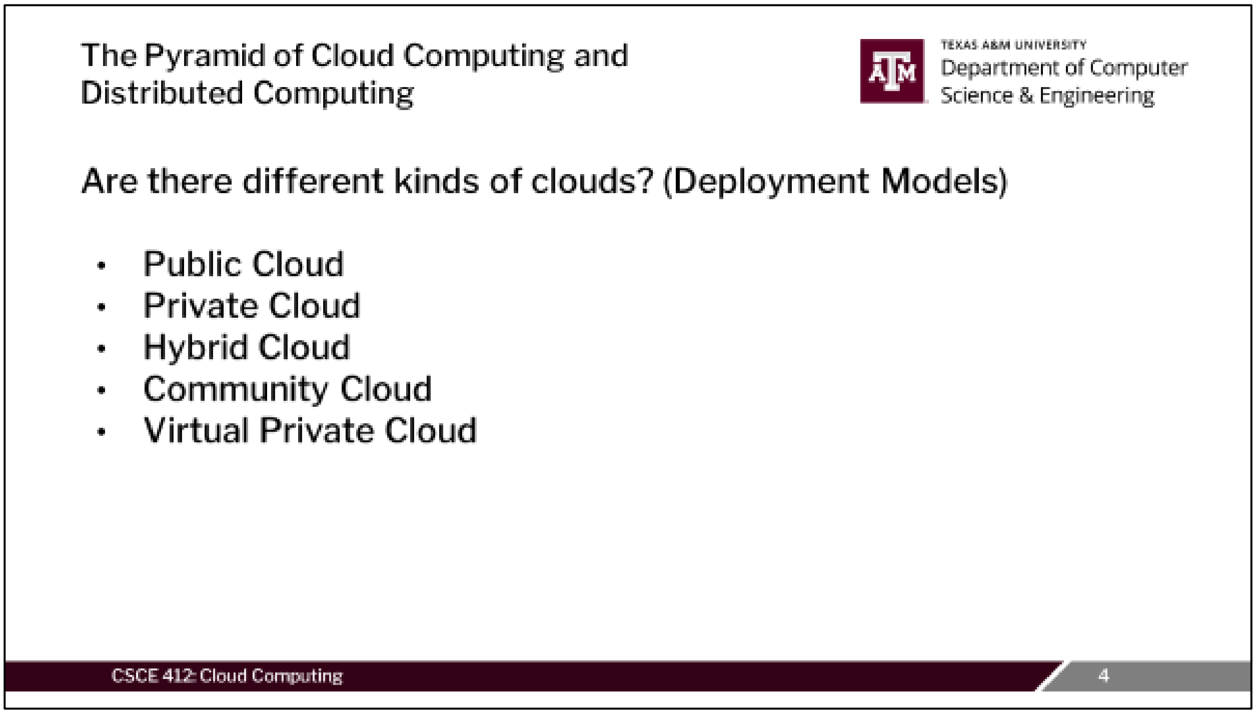
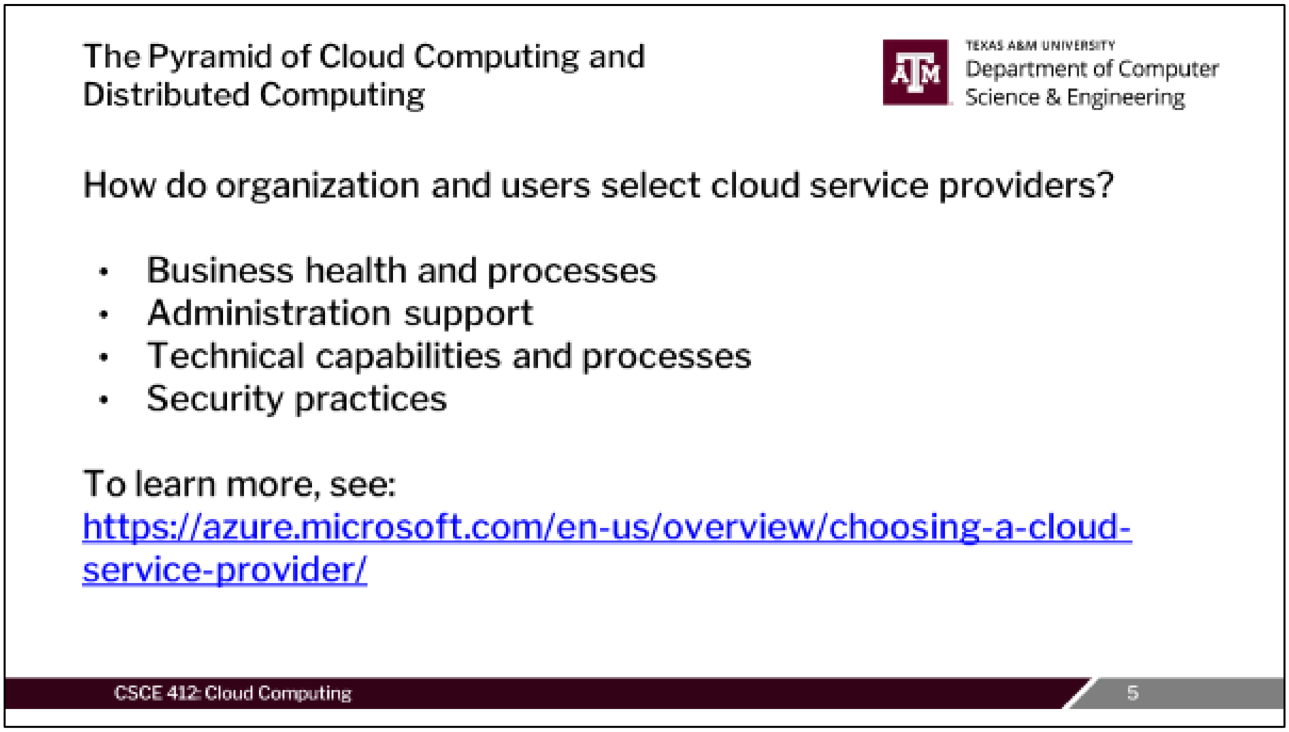


That leads me into the different types, or deployment models, of cloud a user can request. There are five deployment models available to customers - Public, Private, Hybrid, Community, and Virtual Private Clouds. With public clouds, the infrastructure is available to the general public, or really anyone that is able to make digital payments using methods like a credit card and has a very low cost of entry. This means that the same physical resources are split across several different customers. In situations where customers do not want to share physical resources with other customers, they would need to opt for private clouds.

This infrastructure is dedicated to a single customer or organization. This is ideal for customers who want their data, such as health, financial, or other sensitive information, isolated from other tenants. While this type of infrastructure allows organizations to customize the hardware, software, security, and access controls that will be used, it also comes at a significantly higher cost than public cloud models. However, private cloud models may be the only feasible option for some organizations to meet compliance or regulatory standards. In cases where cost optimizations are important and on certain data is sensitive, the hybrid cloud model may suffice. The hybrid cloud is a combination of public cloud and private cloud services into a single, flexible infrastructure that offers the best of both worlds. This model allows organizations to offload computation workloads to appropriate environments while maintaining scalability and security. This is because the hybrid model can move sensitive workloads to the private environment while using the public cloud to handle other tasks. In instances where there are multiple organizations with the same needs and requirements, like financial institutions and healthcare providers, there is an option called community cloud. This model provides pooled resources, like the public cloud, for these groups of similar organizations. The final cloud model is the virtual private cloud, which is when a private cloud model runs on a public cloud infrastructure. With this model, the private cloud is remotely hosted by a cloud provider while retaining the data isolation of a private cloud. To achieve the isolation level on the public cloud, providers implement technologies like subnets, VLANs, and VPNs to segment core networking operations from the public cloud.

Slide 5



Once the organization has selected the type of model to utilize, they must decide what provider to work with. This often takes the form of negotiating service level agreements, or SLAs, which determine the average and/or minimum level of service that the provider must maintain. To learn more, see: https://azure.microsoft.com/en-us/overview/choosing-a-cloud-service-provider/

Slide 6

The Pyramid of Cloud Computing and Distributed Computing

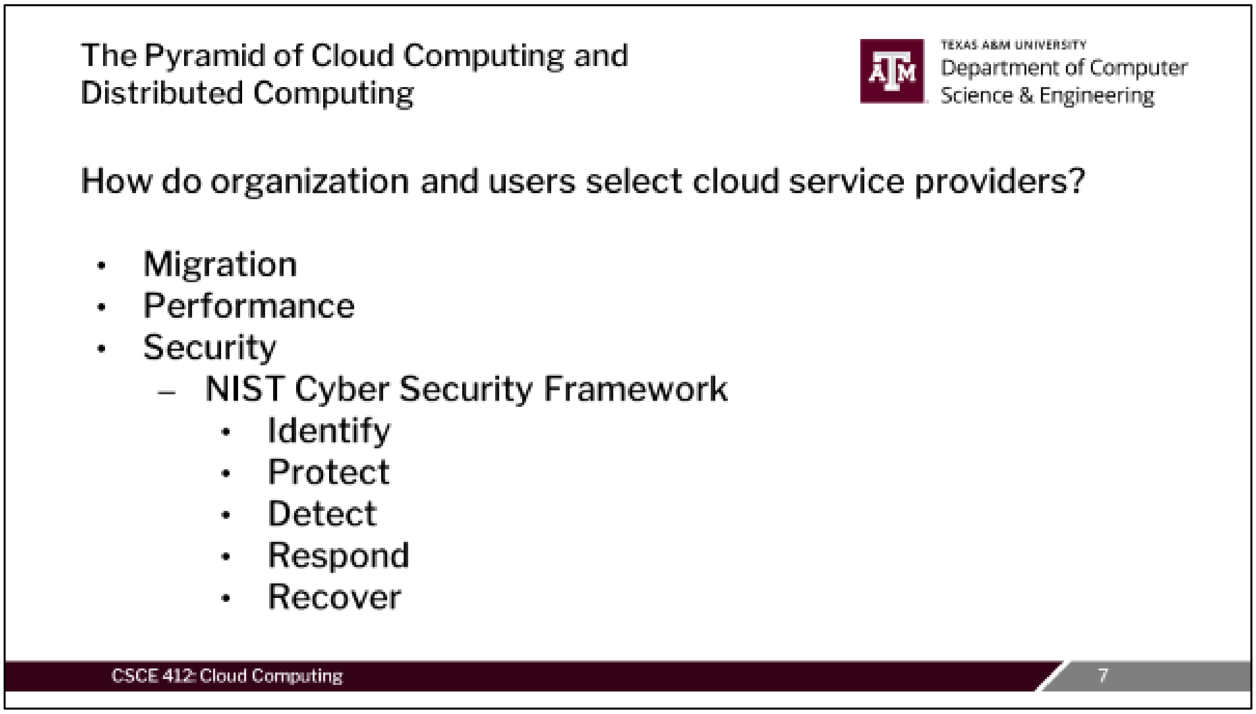
TEXAS A&M UNIVERSITY
Department of Computer Science & Engineering

How do organization and users select cloud service providers?

- Service Level Agreements
  - Response Time
  - Output Bandwidth
  - Number of active servers to monitor for SLA violations
  - Changes in the environment
  - Responding appropriately to guarantee quality of service

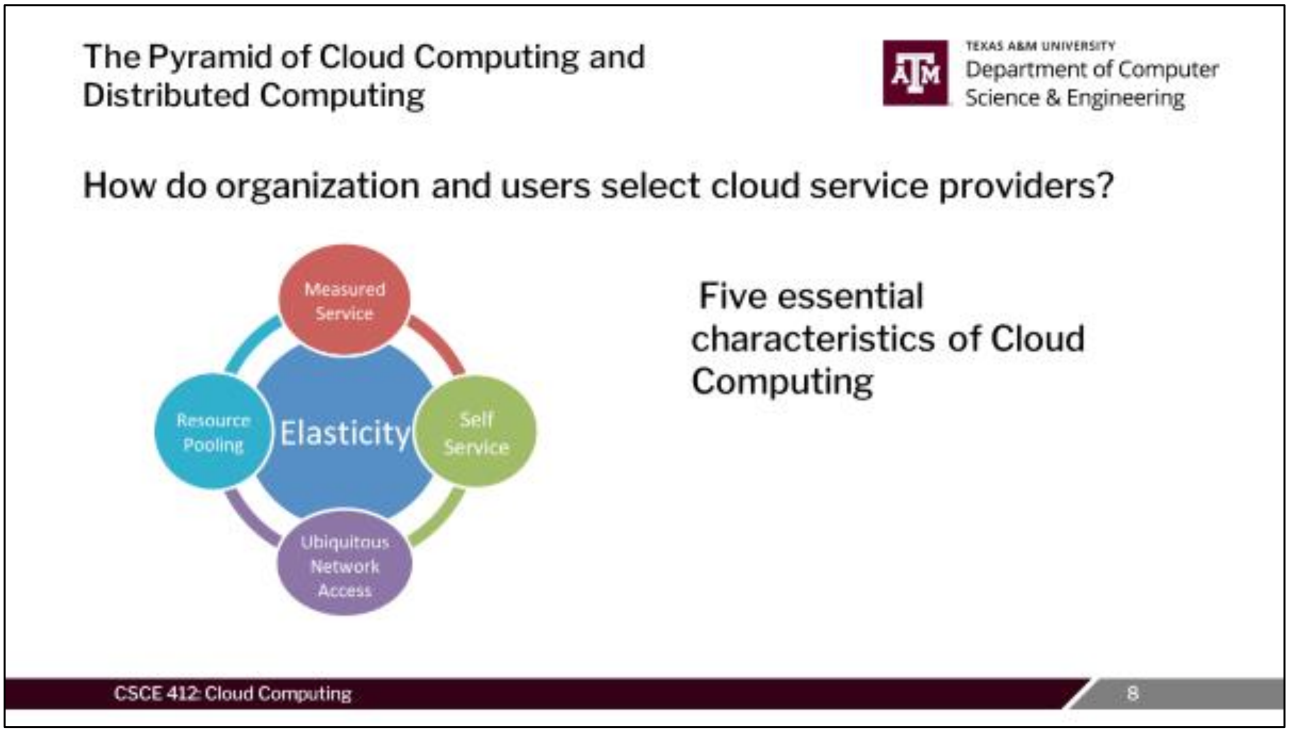
These are attributes such as response time, bandwidth, number of active servers to monitor, changes in the environment, and responding appropriately to guarantee the quality of service. In addition, providers will also demonstrate migration capabilities, system performance, and security practices.

Slide 7



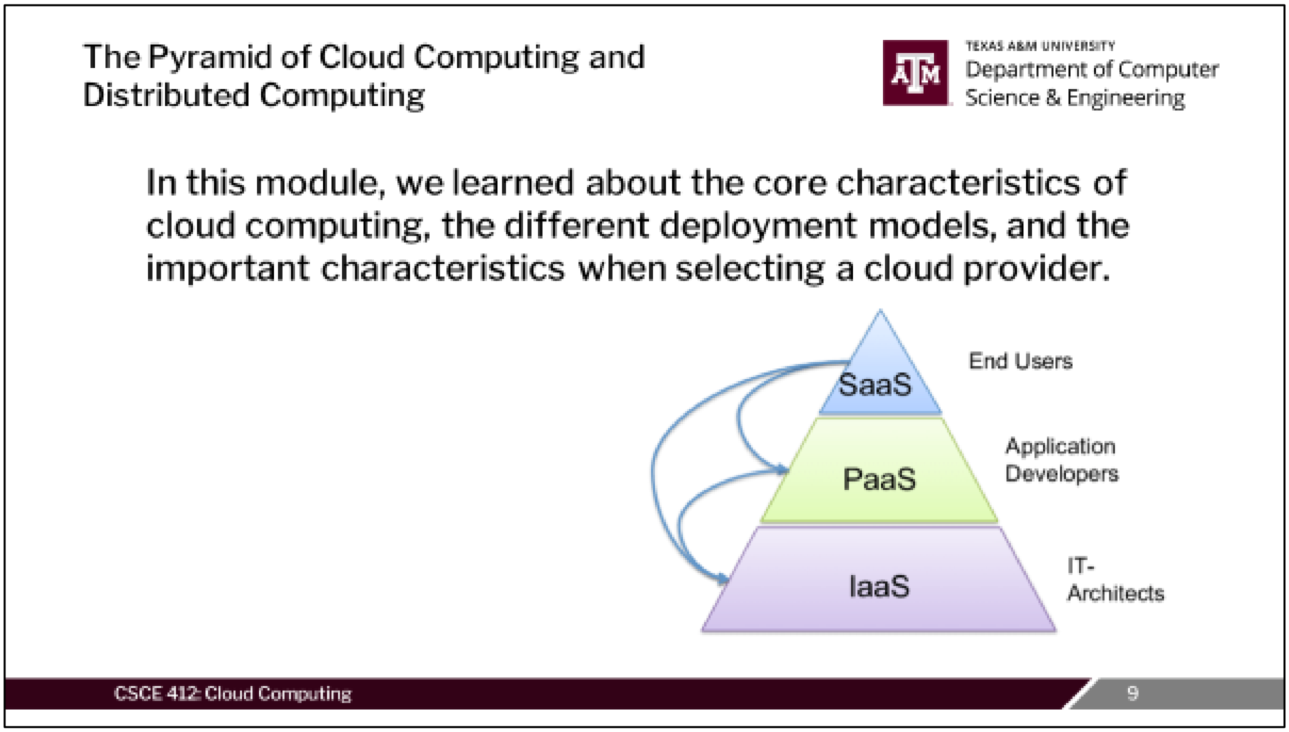
Key considerations are the migration to the cloud, performance of the cloud for their application, and the security in place.

Slide 8



In assessing these qualities, an organization is able to better understand which providers will be able to meet and exceed their needs.

Slide 9

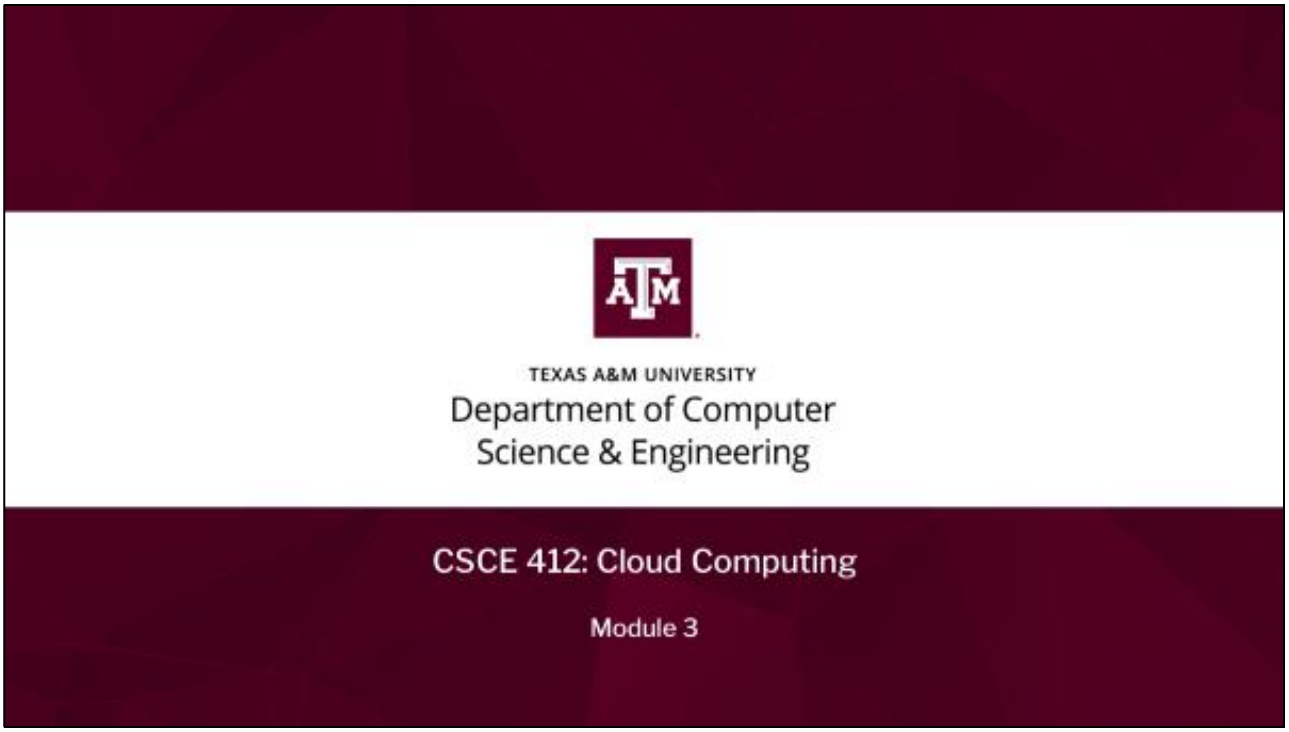


In this module, we learned about the core characteristics of cloud computing, the different deployment models, and the important characteristics when selecting a cloud provider.

Resources:

https://www.ibm.com/cloud/learn/virtualization-a-complete-guide
https://www.ibm.com/cloud/blog/5-benefits-of-virtualization
https://www.ibm.com/cloud/learn/iaas-paas-saas

Slide 10


TEXAS A&M UNIVERSITY
Department of Computer Science & Engineering
CSCE 412: Cloud Computing
Module 3